

ECS Closed-Loop Architecture V3

Goal: establish whether a cost-adjusted intraday edge exists on manifest-verified NSE data. Paper research only; no live-trading approval.

Loop	Setpoint	Measurement	Output	Hard boundary
Entry quality	Cost-aware break-even probability	Completed earlier symbol/side outcomes	Entry derate 0 to 1	Never raises risk
Trade path	Frozen per-trade R path	Current R on completed held bars	One-way stop ratchet	Never loosens stop
Portfolio risk	Heat and drawdown budget	Exposure and drawdown	New-risk derate/halt	Never forces deployment

Data flow

Completed bars -> features -> candidate -> entry-quality feedback -> paper trade -> trade-path feedback -> protective exit -> portfolio feedback -> completed outcome -> later candidates only.

Promotion rule

A fixed alpha needs at least 30 resolved candidates and positive gross and net results in train, validation and untouched test. Passing permits only a separate closed-loop shadow test.